

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE ON COMPOSITION CONTROL, CONTINUOUS POLYMERIZATION, AND POST-POLYMERIZATION MODIFICATION

1. ABSTRACT

This report synthesizes patent evidence relevant to low-acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile incorporation is controlled during copolymerization and how downstream processing affects performance. The target material is a butadiene/acrylonitrile copolymer with acrylonitrile content below standard NBR grades, typically in the low-ACN regime where monomer feed ratio, conversion profile, and polymerization mode strongly influence final composition. Across the selected patents, the landscape spans three main technical directions: (1) direct NBR synthesis in emulsion or solution using controlled monomer feeds and solvent systems, (2) continuous or staged-feed polymerization strategies to manage composition drift and bound acrylonitrile content, and (3) post-polymerization modification of NBR by metathesis or hydrogenation, which is relevant because low-ACN feedstocks are often chosen to balance oil resistance, flexibility, and processability in the final rubber or hydrogenated derivative.

The most composition-relevant disclosures are the continuous/staged polymerization patents, especially those teaching incremental butadiene addition and multi-vessel continuous operation to maintain a high and more uniform bound acrylonitrile content while avoiding premature butadiene depletion. One patent discloses a continuous process with an initial acrylonitrile/butadiene charge of about 80/20 parts and maintenance of unconverted monomers in the range of about 70–90 parts acrylonitrile to 30–10 parts butadiene. Another teaches batch and continuous incremental addition strategies, including 38 parts acrylonitrile with 62 parts butadiene, 5 °C polymerization, and 90% conversion termination. A solution-polymerization patent emphasizes use of at least two solvents, with the main solvent at 70–99.9 vol%, and mercaptan regulators to improve time-conversion behavior.

The remaining patents are indirectly relevant: metathesis catalysts and catalyst-salt systems enable molecular-weight reduction of NBR prior to hydrogenation, while hydrogenation catalyst patents focus on preserving molecular weight during conversion to HNBR. Together, these disclosures show that low-ACN NBR synthesis is most effectively achieved by feed-ratio control, staged or continuous monomer addition, low-temperature free-radical polymerization,

mercaptan-based molecular-weight regulation, and careful coagulation/hydrogenation handling to preserve polymer quality.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9605088B2
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber using a ruthenium- or osmium-based catalyst composition formed by pre-contacting the complex catalyst with hydrogen before exposure to the NBR feed. The process is carried out in organic solvent under hydrogen pressure and is designed to hydrogenate the diene unsaturation while minimizing metathesis-driven molecular-weight loss. - Catalyst pretreatment temperature: 75 to 200° C. — Claim 1 - Hydrogen pressure: 600 psi — Examples 1-6 - Reaction temperature: 120° C. — Examples 1-6 - Agitation speed: 800 rpm — Examples 1-6 - Hydrogen pretreatment temperature: 70° C. — Example 4 - Hydrogen pretreatment time: 12 hours — Example 4 - Hydrogen pretreatment temperature: 120° C. — Example 5 - Hydrogen pretreatment time: 12 hours — Example 5 - Hydrogen pretreatment temperature: 100° C. — Example 6 - Hydrogen pretreatment time: 12 hours — Example 6 - Hydrogenation time: 4 hours — Examples 2, 4, 5 - Hydrogenation time: 5 hours — Example 1 - Hydrogenation time: 18 hours — Example 3 - Hydrogenation time: 22 hours — Example 6 - Perbunan 3431VP concentration: 6 wt % — Examples 1, 3, 4, 5 - Perbunan 3431VP concentration: 3 wt % — Example 2 - Perbunan 3431VP mass: 18 g — Examples 1, 3, 4, 5 - Perbunan 3431VP mass: 9 g — Example 2 - Perbunan 3431VP mass: 36 g — Example 6 - MCB mass: 282 g — Examples 1, 3, 4, 5 - MCB mass: 291 g — Example 2 - MCB mass: 564 g — Example 6 - Wilkinson's catalyst: 15 mg — Example 1 - PPh₃: 18 mg — Example 1 - Zhan 1B: 9 mg — Example 2 - Zhan 1B: 18 mg — Example 3 - Zhan 1B: 48 mg — Example 4 - Catalyst (2): 18 mg — Example 5 - Catalyst (1): 36 mg — Example 6 - Catalyst (2): 48 mg — Example 4 - Hydrogenation degree: 90.3% — Example 1 - Hydrogenation degree: 98.6% — Example 2 - Hydrogenation degree: 92.6% — Example 3 - Hydrogenation degree: 98.9% — Example 4 - Hydrogenation degree: 99.0% — Example 5 - Hydrogenation degree: 97.7% — Example 6 - Mn: 76,286 — Example 1 - Mw: 260,572 — Example 1 - PDI: 3.42 — Example 1 - Mn: 5,560 — Example 2 - Mw: 14,407 — Example 2 - PDI:

2.59 — Example 2 - Mn: 10,103 — Example 3 - Mw: 19,964 — Example 3 - PDI: 1.98 — Example 3 - Mn: 11,742 — Example 4 - Mw: 24,438 — Example 4 - PDI: 2.08 — Example 4 - Mn: 77,726 — Example 5 - Mw: 219,136 — Example 5 - PDI: 2.82 — Example 5 - Mn: 67,546 — Example 6 - Mw: 175,333 — Example 6 - PDI: 2.60 — Example 6

Relevant Experimental Evidence - Example 1: Wilkinson's catalyst gave 90.3% hydrogenation after 5 hours, but with substantial molecular-weight retention issues. - Example 2: Zhan 1B without hydrogen pretreatment achieved 98.6% hydrogenation in 4 hours but produced much lower molecular weight HNBR. - Example 4: Hydrogen pretreatment at 70° C. improved hydrogenation to 98.9% while still causing strong molecular-weight loss. - Example 5: Hydrogen pretreatment at 120° C. gave 99.0% hydrogenation and preserved Mn near the feedstock level. - Example 6: Catalyst (1) with hydrogen pretreatment at 100° C. gave 97.7% hydrogenation with Mn 67,546 and Mw 175,333.

Technical Relevance This patent is relevant because it addresses the downstream hydrogenation step used to convert NBR into HNBR, a common route for low-ACN NBR-derived materials. It shows how catalyst pretreatment and hydrogenation conditions can preserve molecular weight, which is important when starting from composition-controlled NBR feedstocks.

Patent 2

- Patent Number: US9060566B2
- Patent Title: Shoe sole comprising a rubber composition based on nitrile-butadiene rubber, an ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a formulation patent for shoe soles based on nitrile-butadiene rubber rather than a synthesis patent. It discloses NBR-containing rubber compositions with a reinforcing filler, a hydrocarbon resin, a liquid plasticizer, and a crosslinking system to improve grip and wear while maintaining low hydrocarbon absorption. - NBR content: 60 to 100 phr — Claim 1 - Reinforcing filler: 10 to 50 phr — Claim 1 - Hydrocarbon-based resin Tg: greater than 20° C. — Claim 1 - Liquid plasticizer Tg: less than -20° C. — Claim 1 - Resin content A: 4 to 30 phr — Claim 3 - Liquid plasticizer content B: 4 to 30 phr — Claim 4 - A+B content: 8 to 40 phr — Claim 5 - NBR nitrile monomer content: 25% to 45% by weight — Detailed description - NBR nitrile monomer content: 10% to 60% — Detailed description - NBR nitrile monomer content: 20% to 50% — Detailed description - NBR nitrile monomer content: 25% to 45% — Detailed description - Perbunam 3445F acrylonitrile content: approximately 34% by weight — Detailed description

Relevant Experimental Evidence - No numbered examples were detected by the parser; the patent primarily defines composition ranges and intended footwear performance.

Technical Relevance This patent is relevant because it explicitly identifies NBR composition windows, including a commercial grade with approximately 34 wt% acrylonitrile. Although not a low-ACN synthesis disclosure, it provides application-side evidence for NBR grades and formulation constraints that motivate composition selection.

Patent 3

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis catalyst systems for degrading nitrile rubber, using a ruthenium- or osmium-based catalyst together with salts such as LiBr or CsBr to accelerate metathesis. The process is intended to reduce NBR molecular weight before subsequent hydrogenation to HNBR. - Salt: 1 mg LiBr — Example 7.02 - Salt: 1 mg CsBr — Example 8.02 - Grubbs II catalyst: 10 mg — Example 7 - Chlorobenzene: 0.3 mL — Example 7 - CDCl₃: 0.2 mL — Example 7 - Diethyl diallylmalonate: 0.15 mL — Example 7 - Room temperature: room temperature — Example 7 - Reaction time: 30 min — Example 7 - Reaction time: 60 min — Example 7 - Conversion: 21.3% — Example 7.02 at 30 min - Conversion: 57.7% — Example 7.02 at 60 min - Conversion: 55.4% — Example 7.01 at 30 min - Conversion: 100% — Example 7.01 at 60 min - Conversion: 13.4% — Example 8.02 at 15 min - Conversion: 25.3% — Example 8.02 at 30 min - Conversion: 46.5% — Example 8.02 at 60 min - Conversion: 71.9% — Example 8.02 at 90 min - Conversion: 96.2% — Example 8.02 at 150 min - Conversion: 16.5% — Example 8.01 at 15 min - Conversion: 40.3% — Example 8.01 at 30 min - Conversion: 68.9% — Example 8.01 at 60 min - Conversion: 84.7% — Example 8.01 at 90 min - Conversion: 100% — Example 8.01 at 150 min

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis relative to the no-salt control. - Example 8: CsBr also accelerated metathesis, but less strongly than LiBr in the reported comparison.

Technical Relevance This patent is relevant because metathesis is one route to reduce NBR molecular weight before hydrogenation, which can be important for processing low-ACN feedstocks. The salt-accelerated catalyst system offers a way to improve metathesis efficiency and thereby tune NBR molecular weight.

Patent 4

- Patent Number: US8362154B2
- Patent Title: Metathesis of nitrile rubbers in the presence of transition metal complex ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis degradation of nitrile rubber using specific ruthenium catalysts, comparing Grubbs III and Grubbs II systems in the presence of 1-hexene. The goal is to reduce molecular weight and narrow polydispersity of NBR prior to hydrogenation. - NBR mass: 40 g — Examples 1 and 2 - Grubbs III catalyst: 80 mg — Example 1.1 - Grubbs II catalyst: 80 mg — Example 1.2 - Grubbs III catalyst: 200 mg — Example 2.1 - Grubbs II catalyst: 200 mg — Example 2.2 - 1-Hexene: 0.8 g — Examples 1 and 2 - 1-Hexene: 2.0 phr — Examples 1 and 2 - Ruthenium loading: 23 ppm — Example 1 - Ruthenium loading: 57 ppm — Example 2 - Temperature: 23° C. — Examples 1 and 2 - Reaction time: 15 min — Example 1.1 - Reaction time: 60 min — Example 1.1 - Reaction time: 90 min — Example 1.2 - Reaction time: 15 min — Example 2.1 - Reaction time: 90 min — Example 2.1 - Reaction time: 30 min — Example 2.2 - Molecular weight Mw: 240 kg/mol — Example 1.1 and 1.2 initial - Molecular weight Mw: 150 kg/mol — Example 1.1 at 15 min - Molecular weight Mw: 144 kg/mol — Example 1.1 at 60 min - Molecular weight Mw: 204 kg/mol — Example 1.2 at 15 min - Molecular weight Mw: 168 kg/mol — Example 1.2 at 90 min - Molecular weight Mn: 100 kg/mol — Example 1.1 and 1.2 initial - Molecular weight Mn: 71 kg/mol — Example 1.1 at 15 min - Molecular weight Mn: 72 kg/mol — Example 1.1 at 60 min - Molecular weight Mn: 70 kg/mol — Example 1.2 at 15 min - Molecular weight Mn: 62 kg/mol — Example 1.2 at 90 min - PDI: 2.4 — Example 1.1 and 1.2 initial - PDI: 2.1 — Example 1.1 at 15 min - PDI: 2.0 — Example 1.1 at 60 min - PDI: 2.9 — Example 1.2 at 15 min - PDI: 2.7 — Example 1.2 at 90 min - Molecular weight Mw: 39 kg/mol — Example 2.1 at 15 min - Molecular weight Mw: 36 kg/mol — Example 2.1 at 90 min - Molecular weight Mn: 21 kg/mol — Example 2.1 at 15 min - Molecular weight Mn: 20 kg/mol — Example 2.1 at 90 min - PDI: 1.8 — Example 2.1 at 15 min - PDI: 1.8 — Example 2.1 at 90 min - Molecular weight Mw: 176 kg/mol — Example 2.2 at 30 min - Molecular weight Mw: 65 kg/mol — Example 2.2 at 90 min - Molecular weight Mn: 64 kg/mol — Example 2.2 at 30 min - Molecular weight Mn: 33 kg/mol — Example 2.2 at 90 min - PDI: 2.7 — Example 2.2 at 30 min - PDI: 1.9 — Example 2.2 at 90 min

Relevant Experimental Evidence - Example 1: Grubbs III catalyst at 23 ppm Ru gave faster molecular-weight reduction than Grubbs II at the same loading. - Example 2: At 57 ppm Ru, Grubbs III reduced Mw to 36 kg/mol by 90 min and maintained gel-free product. - Example 2: Grubbs II at the same loading was slower and gave higher Mw at comparable times.

Technical Relevance This patent is relevant because it provides a route to lower the molecular weight of NBR feedstocks before hydrogenation, which can be important for processing and for tailoring low-ACN NBR-derived HNBR. It also shows that catalyst choice and co-olefin addition strongly affect degradation efficiency.

Patent 5

- Patent Number: US20140235744A1
- Patent Title: Process for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical emulsion polymerization of acrylonitrile and butadiene in organic solvent mixtures, followed by optional hydrogenation. The process uses at least two solvents, with the main solvent comprising 70 to 99.9 vol% of the total solvent, and may use molar-mass regulators such as mercaptans. - Main solvent fraction: 70 to 99.9% by volume — Claim 1 - Acrylonitrile feed: 38 parts — Synthesis Example 1 - Butadiene feed: 62 parts — Synthesis Example 1 - Acrylonitrile feed: 38 parts — Synthesis Example 2 - Butadiene feed: 62 parts — Synthesis Example 2 - Acrylonitrile feed: 38 parts — Synthesis Example 3 - Butadiene feed: 62 parts — Synthesis Example 3 - Acrylonitrile feed: 38 parts — Synthesis Example 4 - Butadiene feed: 62 parts — Synthesis Example 4 - Water: 200 parts — Synthesis Examples 1-4 - Sodium carbonate: 0.2 part — Synthesis Examples 1-4 - Fatty acid potassium soap: 2.25 part — Synthesis Examples 1-4 - Molecular weight adjuster: 0.5 part — Synthesis Examples 1, 2, 4 - Molecular weight adjuster: 0.1 part — Synthesis Example 3 - Reaction temperature: 5° C. — Synthesis Examples 1-4 - Cumen hydroperoxide: 0.1 part — Synthesis Examples 1-4 - Polymerization conversion rate: 90% — Synthesis Examples 1-4 - Hydroquinone terminator: 0.1 part of 10 wt % solution — Synthesis Examples 1-4 - Post-polymerization stirring: 30° C. for 3 hours — Synthesis Examples 1-4 - KOH solution: 2.5 wt % — Synthesis Examples 1-4 - KOH addition: 0.5 part — Synthesis Examples 1-4 - Concentration step: 60° C. water temperature — Synthesis Examples 1-4 - Residual organic acid: 7 wt % — Example 1, 2, 3 - Residual organic acid: 25 wt % — Comparative Example 5 - Residual organic acid: 30 wt % — Comparative Example 6 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Example 1 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Example 4 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Example 6 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Comparative Example 4 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Comparative Example 5 - Hydrogenation catalyst: 200 wt ppm palladium-silica — Comparative Example 6 - Mercapto compound: 20 wt ppm — Example 1 - Mercapto compound: 80 wt ppm — Example 2 - Mercapto compound: 20 wt ppm 2,2,4,6,6-heptamethyl-4-nonanethiol — Example 3 - Mercapto compound: 5 wt ppm — Example 4 - Mercapto compound: 50 wt ppm — Example 5 -

Mercapto compound: 20 wt ppm — Example 6 - Hydrogen pressure: 3.0 MPa — Examples 1, 4, 6 and Comparative Examples 4-6 - Hydrogenation time: 6 hours — Examples 1, 4, 6 and Comparative Examples 4-6 - Hydrogenation rate: 95% — Example 1 - Hydrogenation rate: 90% — Example 2 - Hydrogenation rate: 96% — Example 3 - Hydrogenation rate: 90% — Example 4 - Hydrogenation rate: 95% — Example 5 - Hydrogenation rate: 95% — Example 6 - Hydrogenation rate: 70% — Comparative Example 1 - Hydrogenation rate: 65% — Comparative Example 2 - Hydrogenation rate: 75% — Comparative Example 3 - Hydrogenation rate: 60% — Comparative Example 4 - Hydrogenation rate: 80% — Comparative Example 5 - Hydrogenation rate: 75% — Comparative Example 6

Relevant Experimental Evidence - Synthesis Example 1: NBR latex A1 was prepared from 38 parts acrylonitrile and 62 parts butadiene at 5° C. with cumene hydroperoxide and mercaptan regulator. - Synthesis Example 2: t-dodecylmercaptan was used as the molecular-weight adjuster to produce latex A2 under the same 38/62 monomer ratio. - Example 1: Coagulation at 30° C. followed by hydrogenation with 20 wt ppm mercapto compound gave 95% hydrogenation. - Example 2: Increasing the mercapto compound to 80 wt ppm reduced hydrogenation rate to 90%. - Example 4: Latex A2 with 5 wt ppm mercapto compound gave 90% hydrogenation, showing sensitivity to mercaptan content and latex history.

Technical Relevance This patent is highly relevant because it discloses direct NBR synthesis conditions, including a 38/62 acrylonitrile/butadiene feed ratio, low-temperature emulsion polymerization, and mercaptan-based molecular-weight control. These are core levers for producing low-ACN NBR with controlled composition and processability.

Patent 6

- Patent Number: US5770660A
- Patent Title: Continuous polymerization process for producing NBR rubber having a high bound ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses continuous emulsion polymerization of acrylonitrile and butadiene to produce NBR with high bound acrylonitrile content. It uses staged butadiene addition across multiple reactors to maintain a favorable unconverted monomer ratio and avoid the composition drift seen in conventional batch polymerization. - Acrylonitrile feed: about 80 parts — Claim 1 - Butadiene feed: 20 parts $\pm 10\%$ — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to 30 to 10 parts butadiene — Claim 1 - Acrylonitrile charged: 54 parts — Example I - Butadiene charged: 21 parts — Example I - Conversion: 50.2 percent — Example I - Bound acrylonitrile: 59.5 percent by weight

— Example I - Acrylonitrile charged: 54 parts — Example II - Butadiene charged: 37 parts total — Example II - Initial butadiene charge: 21 parts — Example II - Additional butadiene charge: 16 parts — Example II - Conversion: 35 percent — Example II - Conversion: 73.3 percent — Example II - Bound acrylonitrile: 52.6 percent by weight — Example II - Conversion: 73.6 percent — Example III - Bound acrylonitrile: 54.9 percent — Example III - Acrylonitrile charged: 54 parts — Example IV - Butadiene charged: 46 parts total — Example IV - Initial butadiene charge: 21 parts — Example IV - Additional butadiene charge: 16 parts — Example IV - Additional butadiene charge: 9 parts — Example IV - Conversion: 35 percent — Example IV - Conversion: 65 percent — Example IV - Conversion: 85.2 percent — Example IV - Bound acrylonitrile: 50.5 percent — Example IV - Acrylonitrile charged: 56 parts — Example V - Butadiene charged: 43 parts total — Example V - Initial butadiene charge: 22 parts — Example V - Additional butadiene charge: 8 parts — Example V - Additional butadiene charge: 7 parts — Example V - Additional butadiene charge: 6 parts — Example V - Conversion: 32 percent — Example V - Conversion: 52 percent — Example V - Conversion: 72 percent — Example V - Final conversion: 81.9 percent — Example V - Bound acrylonitrile: 50.2 percent — Example V - Final conversion: 86.8 percent — Example VI - Bound acrylonitrile: 50.6 percent — Example VI - Acrylonitrile charged: 56 parts — Example VII - Butadiene charged: 20 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Additional butadiene charge: 6 parts — Example VII - Steady-state conversion: 24 percent — Example VII - Steady-state conversion: 40 percent — Example VII - Steady-state conversion: 56 percent — Example VII - Steady-state conversion: 72 percent — Example VII - Steady-state conversion: 82 percent — Example VII - Bound acrylonitrile: 48.8 percent — Example VII

Relevant Experimental Evidence - Example I: Conventional batch charging gave 59.5 wt% bound acrylonitrile but only 50.2% conversion. - Example II: Two-increment butadiene addition improved conversion to 73.3% and yielded 52.6 wt% bound acrylonitrile. - Example IV: Three butadiene increments improved conversion to 85.2% and produced 50.5% bound acrylonitrile. - Example V: Four increments and 81.9% final conversion gave 50.2 wt% bound acrylonitrile. - Example VII: Continuous multi-vessel operation produced 48.8% bound acrylonitrile with superior property uniformity.

Technical Relevance This is one of the most directly relevant patents because it teaches how to control acrylonitrile incorporation through staged and continuous monomer addition. The disclosed feed ratios and conversion management are central to producing NBR in the low-to-moderate acrylonitrile range with improved uniformity.

Patent 7

- Patent Number: US10781270B2
- Patent Title: Method of production of highly saturated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses production of nitrile rubber followed by coagulation at controlled temperature and hydrogenation in the presence of a palladium-silica catalyst. The key innovation is controlling residual mercapto-compound content in the hydrogenation feed to improve hydrogenation reactivity. - Coagulation temperature: 10 to 60° C. — Claim 1 - Hydrogen conversion rate: 50% or more — Claim 1 - Mercapto compound (9 to 12 carbon atoms): 50 wt ppm or less — Claim 1 - Mercapto compound (13 to 16 carbon atoms): 1 to 120 wt ppm — Claim 1 - Acrylonitrile monomer content: 8 to 60 wt % — Detailed description - Acrylonitrile monomer content: 12 to 58 wt % — Detailed description - Acrylonitrile monomer content: 16 to 50 wt % — Detailed description - Diene monomer content: 40 to 90 wt % — Detailed description - Diene monomer content: 41 to 85 wt % — Detailed description - Diene monomer content: 43 to 80 wt % — Detailed description - Aluminum sulfate: 3 wt % with respect to nitrile rubber — Example 1 - Coagulation temperature: 30° C. — Examples 1-6 - Drying: 60° C. for 12 hours — Examples 1-6 - Acetone concentration: 12 wt % — Examples 1-6 - Palladium-silica catalyst: 200 wt ppm — Examples 1-6 and Comparative Examples 4-6 - 2,2,4,6,6-pentamethyl-4-octanethiol: 20 wt ppm — Example 1 - 2,2,4,6,6-pentamethyl-4-octanethiol: 80 wt ppm — Example 2 - 2,2,4,6,6-heptamethyl-4-nonanethiol: 20 wt ppm — Example 3 - 2,2,4,6,6-pentamethyl-4-octanethiol: 5 wt ppm — Example 4 - 2,2,4,6,6-pentamethyl-4-octanethiol: 50 wt ppm — Example 5 - Hydrogen pressure: 3.0 MPa — Examples 1, 4, 6 and Comparative Examples 4-6 - Hydrogenation time: 6 hours — Examples 1, 4, 6 and Comparative Examples 4-6 - Hydrogenation rate: 95% — Example 1 - Hydrogenation rate: 90% — Example 2 - Hydrogenation rate: 96% — Example 3 - Hydrogenation rate: 90% — Example 4 - Hydrogenation rate: 95% — Example 5 - Hydrogenation rate: 95% — Example 6 - Hydrogenation rate: 70% — Comparative Example 1 - Hydrogenation rate: 65% — Comparative Example 2 - Hydrogenation rate: 75% — Comparative Example 3 - Hydrogenation rate: 60% — Comparative Example 4 - Hydrogenation rate: 80% — Comparative Example 5 - Hydrogenation rate: 75% — Comparative Example 6 - Residual organic acid: 7 wt % — Examples 1-4 - Residual organic acid: 25 wt % — Comparative Example 5 - Residual organic acid: 30 wt % — Comparative Example 6

Relevant Experimental Evidence - Example 1: 20 wt ppm of 13-carbon mercapto compound and 30° C. coagulation gave 95% hydrogenation. - Example 2: Raising the mercapto compound to 80 wt ppm reduced hydrogenation rate to 90%. - Example 3: Using a 16-carbon mercapto compound at 20 wt ppm gave 96% hydrogenation. - Example 4: A latex with 7 wt% residual organic acid and 5 wt ppm mercapto compound gave 90% hydrogenation. - Example 5: Increasing mercapto compound to 50 wt ppm improved hydrogenation to 95%.

Technical Relevance This patent is relevant because it links NBR coagulation history and mercapto-compound residue to hydrogenation efficiency, which matters for downstream processing of low-ACN NBR. It also provides explicit composition windows for the NBR precursor and demonstrates how to preserve hydrogenation reactivity.

Patent 8

- Patent Number: US8536277B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions, including degradation of nitrile rubber, using ruthenium- or osmium-based metathesis catalysts together with added salts. The salt/catalyst combination is intended to increase metathesis activity and improve molecular-weight reduction of NBR. - Salt:catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Hydrated nitrile rubber residual double bond content: 0 to 18% — Detailed description - Hydrated nitrile rubber residual double bond content: not more than about 0.9% for fully hydrated grades — Detailed description - Mooney viscosity: 55 to 120 — Detailed description - Number average molecular weight Mn: about 200,000 to 700,000 — Detailed description - NBR feedstock Mooney viscosity: about 30 Mooney units — Detailed description - NBR feedstock Mooney viscosity: less than 30 Mooney units — Detailed description - NBR feedstock Mn: less than 70,000 g/mol — Detailed description - Hydrated nitrile rubber Mw: 30,000 to 250,000 — Detailed description - Hydrated nitrile rubber Mooney viscosity: 3 to 50 — Detailed description - PDI: less than 2.5 — Detailed description - Grubbs I catalyst: 0.5 phr — Detailed description - Ruthenium content: 614 ppm — Detailed description - Ruthenium content: 307 ppm — Detailed description - Ruthenium content: 61 ppm — Detailed description - Conversion with $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$: 68.5% — Example discussion - Conversion with SnBr_2 : 71.9% — Example discussion - Conversion without salt: 25.8% — Example discussion - Conversion with SnI_2 : 4.1% — Example discussion

Relevant Experimental Evidence - No numbered examples were detected by the parser; the patent mainly provides catalyst-system definitions and literature comparisons. - The description highlights that salt addition can increase metathesis activity, but the effect depends strongly on the salt identity.

Technical Relevance This patent is relevant because it provides a route to reduce NBR molecular weight by metathesis, a useful precursor step for processing and hydrogenation of low-ACN NBR-derived materials. The salt-boosted catalyst system offers a way to improve efficiency at lower catalyst loadings.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5770660A and US20140235744A1 are the most composition-direct patents: both teach free-radical copolymerization of acrylonitrile and butadiene, but US5770660A emphasizes continuous staged-feed operation while US20140235744A1 emphasizes solvent-mixture solution polymerization and mercaptan regulation.
- US5770660A uses high acrylonitrile feed ratios and incremental butadiene addition to maintain bound acrylonitrile near 48.8-59.5 wt%, whereas US20140235744A1 uses a 38/62 acrylonitrile/butadiene feed in emulsion/solution processing and focuses on processability and hydrogenation compatibility rather than maximizing bound ACN.
- US10781270B2 and US9605088B2 are downstream hydrogenation patents; both use organic-medium hydrogenation of NBR, but US10781270B2 focuses on mercapto-residue control and coagulation temperature, while US9605088B2 focuses on catalyst pretreatment to suppress metathesis and preserve molecular weight.
- US8362154B2 and US8536277B2 are metathesis/degradation patents; both target molecular-weight reduction of NBR before hydrogenation, but US8362154B2 compares Grubbs III vs Grubbs II with 1-hexene, while US8536277B2 adds salts to boost catalyst activity.
- US8957161B2 and US8536277B2 both use salt effects to enhance metathesis, but US8957161B2 specifically demonstrates LiBr and CsBr acceleration in model metathesis and discusses nitrile-rubber degradation, whereas US8536277B2 claims a broader salt:catalyst ratio window and includes tin-salt examples.
- US9060566B2 is formulation-focused rather than synthesis-focused; it confirms practical NBR composition windows in footwear and cites a commercial NBR with approximately 34 wt% acrylonitrile, which is above the low-ACN target but useful as an application benchmark.
- Across the set, the most relevant synthesis levers for low-ACN NBR are monomer feed ratio, staged or continuous butadiene addition, low-temperature free-radical polymerization, mercaptan-based molecular-weight control, and careful coagulation/hydrogenation conditions to preserve polymer quality.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR is best approached through controlled free-radical copolymerization of acrylonitrile and butadiene, with composition governed primarily by feed ratio, staged addition, and conversion management. The strongest direct synthesis guidance comes from continuous/staged-feed patents, which show that maintaining a high acrylonitrile-to-butadiene ratio in the unconverted monomer pool and using incremental butadiene addition can stabilize composition and improve polymer properties. For practical low-ACN production, the most defensible parameter set from the evidence is low-temperature emulsion or solution polymerization near 5° C., acrylonitrile/butadiene feeds in the approximate 38/62 to 80/20 range depending on target composition, mercaptan or related molecular-weight regulators at low ppm to sub-part levels, and coagulation in the 10-60° C. range before any hydrogenation step. If downstream hydrogenation is required, the data support careful control of residual mercapto compounds and catalyst pretreatment to avoid molecular-weight loss and maintain reactivity. Overall, the patent set suggests that composition control during polymerization is the key

determinant for low-ACN NBR, while metathesis and hydrogenation patents provide enabling downstream processing routes.

5. REFERENCES

- US9605088B2 | Catalyst compositions and their use for hydrogenation of nitrile rubber | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9605088B2/en>
- US9060566B2 | Shoe sole comprising a rubber composition based on nitrile-butadiene rubber, an ... | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9060566B2/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US8362154B2 | Metathesis of nitrile rubbers in the presence of transition metal complex ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8362154B2/en>
- US20140235744A1 | Process for producing nitrile rubbers in organic solvents | Not disclosed | US | 2014 | <https://patents.google.com/patent/US20140235744A1/en>
- US5770660A | Continuous polymerization process for producing NBR rubber having a high bound ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5770660A/en>
- US10781270B2 | Method of production of highly saturated nitrile rubber | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10781270B2/en>
- US8536277B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8536277B2/en>